

Vježba 13

Vremensko –brojački
podsustav

Upotreba timera/prekida - Arduino Uno

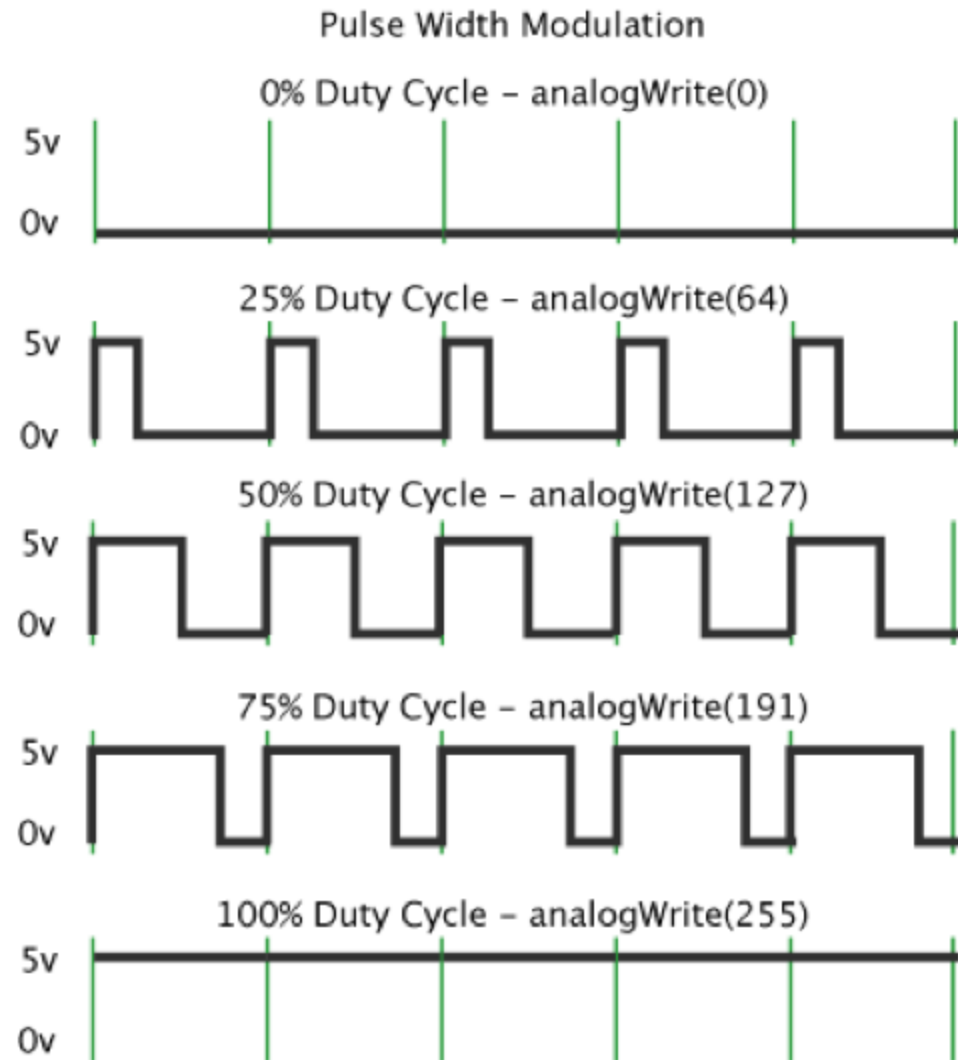
- Odabir vanjskog izvora takta (npr. 2 Hz)
 - CPU 16MHz
 - Maximalna vrijednost registra (8bit $\rightarrow$ 256, 16bit $\rightarrow$ 65536)
 - Takt CPU frekvencije dijeli se sa vrijednosti pre-scalara ($16000000/256=62500$)
 - Podijeliti rezultat s željenom frekvencijom ($62500/2\text{Hz}=31250$)
 - Provjeriti dali vrijednost do koje je potrebno brojiti manja od maksimalne vrijednosti ($31250 < 65536$)

PWM modulacija

- Pulse Width Modulation
 - Kod kontroliranja jačine svijetlosti LED diode ili brzine motora
 - Koristi jedan vodič za simulacija višebitnog broja (0-255)
 - Određivanjem oblika impulsa simulira se visina izlaznog napona

Duty cycle

- Vrijeme trajanja visokog stanja na izlazu
- Value 0 → 0% duty cycle
- Value 127 → 50% duty cycle
- Value 255 → 100% duty cycle



Duty cycle

- Formula

- $\text{Duty_Cycle} = [\text{ON_time} / (\text{ON_time} + \text{OFF_time})] * 100$
 - ON_time → vrijeme stanja HIGH
 - OFF_time → vrijeme stanja LOW
 - Duty cycle → stanje HIGH u postocima

- Primjer

- ON_time → 1 ms
- OFF_time → 1 ms
- Duty cycle → 50%

Duty cycle

- Zadatak
 - odredi duty cycle ako je zadano
 - ON_time → 1 ms
 - OFF_time → 3 ms

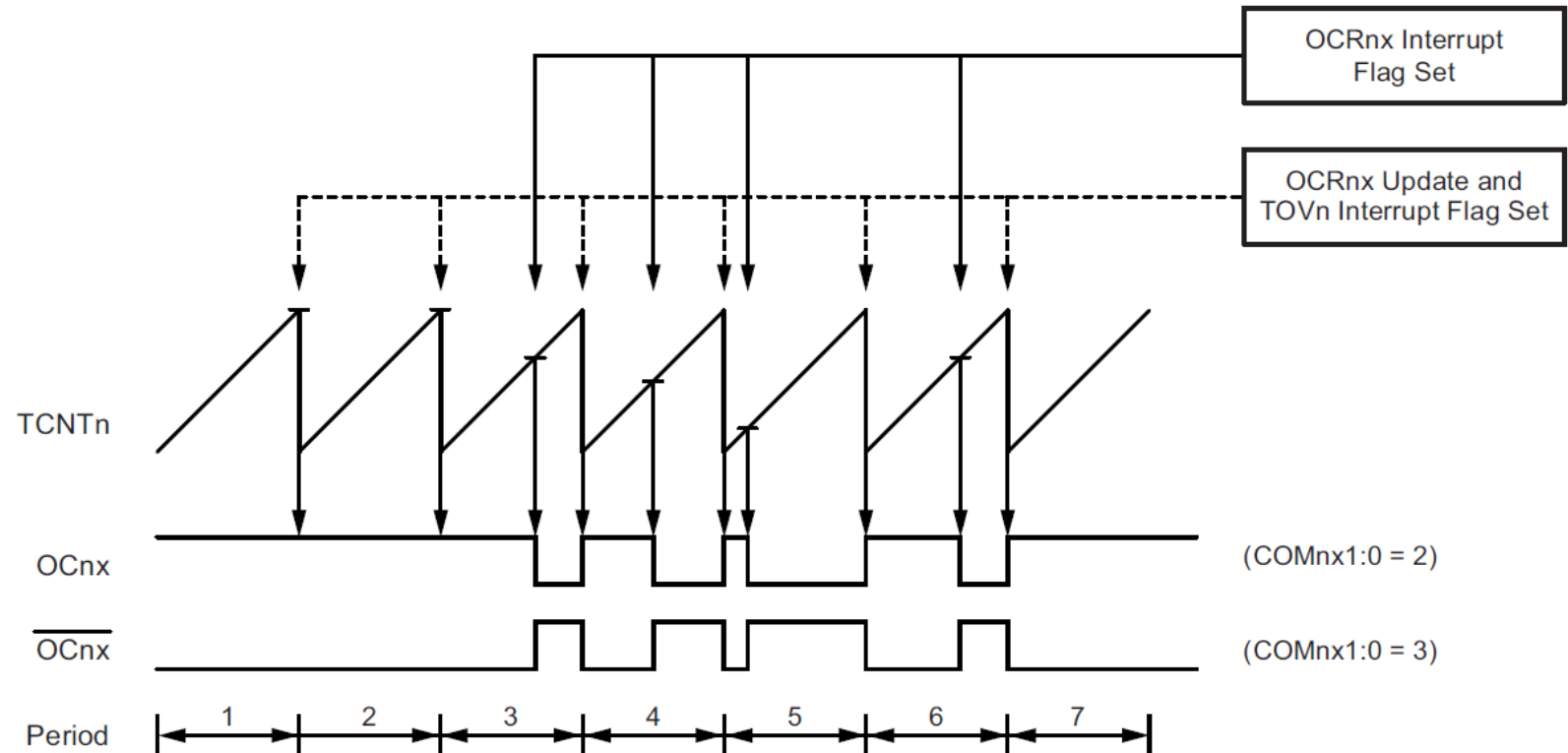
RJEŠENJE → 25 %

Vrijednost izlaznog napona

- Formula
 - $\text{Output_Voltage} = \text{Duty_Cycle} * \text{Input_Voltage}$
 - Input_Voltage → vrijednost ulaznog napona (napajanja)
 - Output_Voltage → vrijednost izlaznog napona
- Primjer
 - Input_Voltage = 5V
 - Duty cycle = 25%
 - Output_Voltage = 1.25V

Fast PWM mode

- Pogodan za
 - Regulaciju snage, pojačanja (upravljanje brzinom motora, jačinom svijetlosti LED diode)



Primjer 2

```
.def temp = R16
.def delay_0 = R17
.def delay_1 = R18
.def counter = R19
init:
LDI temp,0xFF
OUT DDRD,R16
LDI temp,1<<WGM00|1<<WGM01|1<<COM0A1; Fast PWM
(Top OCRA) // Clear OC0A on Compare match -->
PD6
OUT TCCR0A,temp
LDI temp,1<<CS01 ;Clock Select divide by 8
OUT TCCR0B,temp
ldi counter,0x0
loop:
ldi delay_0,228;set delay with values for 10ms
```

```
cycle
ldi delay_1,200
OUT OCR0A,counter
INC Counter
RCALL delay
rjmp loop

delay:
dec delay_0
brne delay
dec delay_1
brne delay
ret
```

Zadatak 2.

Izradi program koji na izlazu PD5 upravlja LED diodom na sljedeći način:

- 2 sekunde ugašena – 2 sekunde upaljenja na 50% intenziteta - 2 sekunde upaljena

Zadatak 2 - rješenje

```
.include "../m328Pdef.inc"

.def temp = R16
.def delay_0 = R17
.def delay_1 = R18
.def counter = R19

init:
LDI temp,0xFF
OUT DDRD,R16

LDI temp,1<<WGM00|1<<WGM01|1<<COM0A1; Fast PWM (Top OCRA) // Clear OC0A on Compare match --> PD6
OUT TCCR0A,temp

LDI temp,1<<CS01 ;Clock Select divide by 8
OUT TCCR0B,temp

ldi counter,0x0

loop:
LDI counter,0
OUT OCR0A,counter
RCALL delay
RCALL delay

LDI counter,125
OUT OCR0A,counter
RCALL delay
RCALL delay
LDI counter,255
```

```
OUT OCR0A,counter
RCALL delay
RCALL delay
rjmp loop

delay:
LDI R16, 127;1 strojni ciklus

Delay1:
LDI R17, 125;1 strojni ciklus

Delay2:
LDI R18, 250 ; 1 strojni ciklus

Delay3:
DEC R18 ; 1 strojni ciklus
NOP ; 1 strojni ciklus
BRNE Delay3 ; 2 strojni ciklus

DEC R17 ; 1 strojni ciklus
BRNE Delay2 ; 2 strojni ciklus

DEC R16 ; 1 strojni ciklus
BRNE Delay1 ; 2 strojni ciklus

ret
```

Zadatak 3.

- Programiraj mikroupravlja da izvršava postupno prijelazno paljenje i gašenje LED diode (dimming)

Rješenje

```
.def temp = R16
.def delay_0 = R17
.def delay_1 = R18
.def counter = R19

init:
    LDI    temp, 0xFF
    OUT    DDRD,temp
    LDI    temp, 1<<WGM00 | 1<<WGM01 | 1<<COM0A1
    OUT    TCCR0A,temp
    LDI    temp, 1<<CS01
    OUT    TCCR0B,temp
    ldi    counter, 0x0

loop:
    count_up:
        ldi    delay_0, 228    ;set delay with values for 10ms cycle
        ldi    delay_1, 104
        OUT    OCR0A, counter
        INC    Counter
```

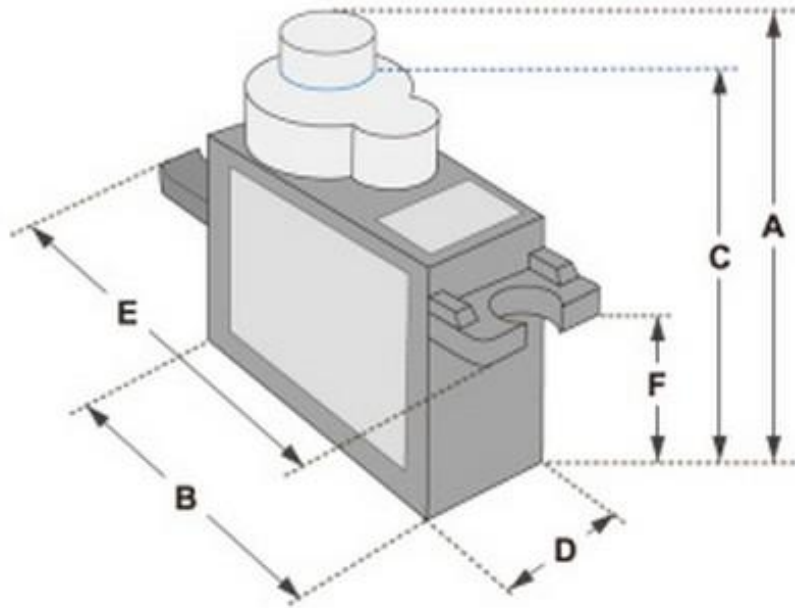
```
RCALL    delay
    cpi    counter, 0xFF
    brne    count_up
count_down:
    ldi    delay_0, 228    ;set delay with values for 10ms cycle
    ldi    delay_1, 104
    OUT    OCR0A, counter
    DEC    counter
    RCALL    delay
    cpi    counter, 0x0
    brne    count_down
    rjmp    loop
delay:
    dec    delay_0
    brne    delay
    dec    delay_1
    brne    delay
    ret
```

Servomotor + timer1 (16
bita)

Servomotor (Servo sg90) + timer1 (16 bita)

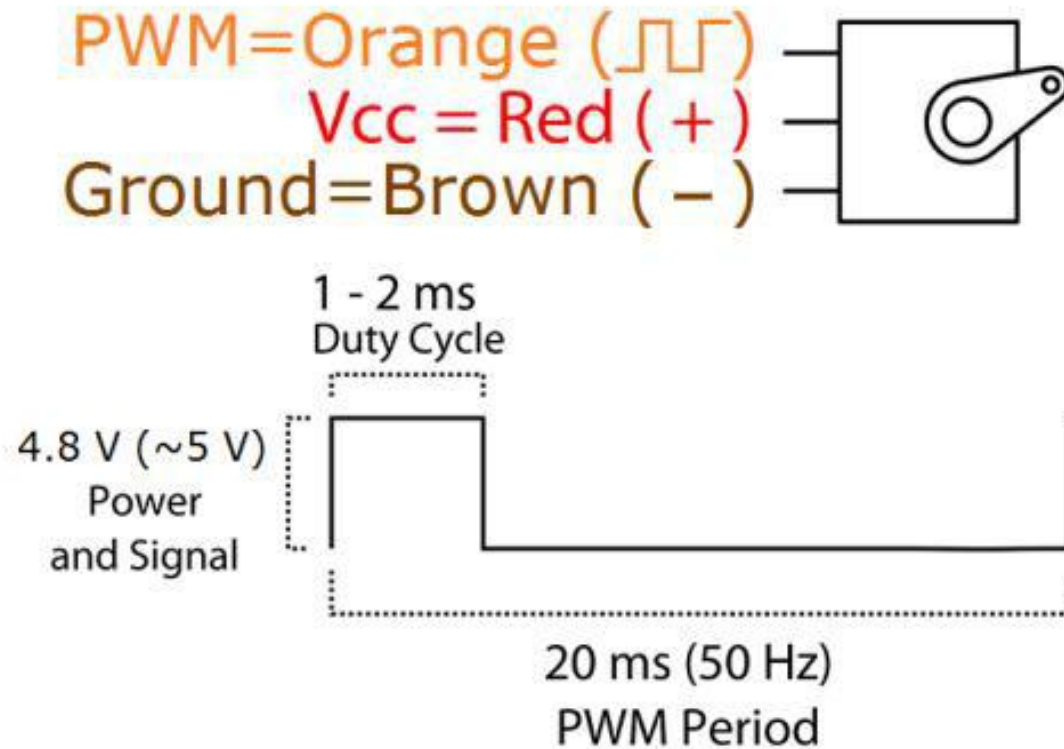


Specifikacije



Dimensions & Specifications
A (mm) : 32
B (mm) : 23
C (mm) : 28.5
D (mm) : 12
E (mm) : 32
F (mm) : 19.5
Speed (sec) : 0.1
Torque (kg-cm) : 2.5
Weight (g) : 14.7
Voltage : 4.8 - 6

Pinovi + upravljački signal



- 3 karakteristične pozicije
 - Pozicija „0” → impuls širine 1.5ms
 - Pozicija „90” → impuls širine 2ms „u desno”
 - Pozicija „-90” → impuls širine 1ms „u lijevo”

Primjer upravljanja (postavljanja u određenu poziciju)

```
.def temp = R16
.def delay_0 = R17
.def delay_1 = R18
.def counter = R19

init:
    LDI    temp, 0xFF
    OUT    DDRB, R16
    LDI    temp, 0xFF
    OUT    DDRD, R16
    LDI    temp, 1<<WGM11 | 1<<COM1A1 |
1<<COM1A0
    sts    TCCR1A, temp
    LDI    temp, 1<<WGM12 | 1<<WGM13 |
1<<CS12 | 0<<CS10
    sts    TCCR1B, temp
    LDI    temp, 0x04
                                sts    ICR1H, temp
                                LDI    temp, 0xE2
                                sts    ICR1L, temp
                                ldi    counter, 0

loop:
                                sts    OCR1AL, counter
                                RCALL  delay
                                rjmp   loop

delay:
    dec    delay_0
    brne   delay
    dec    delay_1
    brne   delay
    ret
```

Zadatak 4.

- Programiraj mikroupravljač da se okreće puni krug u lijevu stranu ,
zatim puni krug u desnu stranu.

Rješenje

```
.include "A328pdef.inc"
```

```
.def temp = R16
```

```
.def delay_0 = R17
```

```
.def delay_1 = R18
```

```
.def counter = R19
```

```
init:
```

```
    LDI    temp, 0xFF
```

```
OUT    DDRB, R16
```

```
LDI    temp, 0xFF
```

```
OUT    DDRD, R16
```

```
    LDI    temp, 1<<WGM11 | 1<<COM1A1 |  
1<<COM1A0
```

```
    sts    TCCR1A, temp
```

```
    LDI    temp, 1<<WGM12 | 1<<WGM13 | 1<<CS12 |  
0<<CS10
```

```
    sts    TCCR1B, temp
```

```
LDItemp,0x04
```

```
    stslCR1H, temp
```

```
LDItemp,0xE2
```

```
stslCR1L, temp
```

```
    ldi    counter,0
```

```
loop:
```

```
count_up:
```

```
    ldi    delay_0, 228    ;set delay with values for  
10ms cycle
```

```
    ldi    delay_1, 204
```

```
    sts    OCR1AL, counter
```

```
    INC    counter
```

```
    RCALL  delay
```

```
    cpi    counter, 0xAF
```

```
    brne   count_up
```

```
count_down:
```

```
    ldi    delay_0, 228    ;set delay with values for  
10ms cycle
```

```
    ldi    delay_1, 204
```

```
    sts    OCR1AL, counter
```

```
    DEC    counter
```

```
    RCALL  delay
```

```
    cpi    counter, 0x50
```

```
    brne   count_down
```

```
    rjmp   loop
```

```
delay:
```

```
    dec    delay_0
```

```
    brne   delay
```

```
    dec    delay_1
```

```
    brne   delay
```

```
ret
```

ATMega328P and Arduino Uno Pin Mapping

Arduino function

reset

digital pin 0 (RX)

digital pin 1 (TX)

digital pin 2

digital pin 3 (PWM)

digital pin 4

VCC

GND

crystal

crystal

digital pin 5 (PWM)

digital pin 6 (PWM)

digital pin 7

digital pin 8

(PCINT14/RESET) PC6

(PCINT16/RXD) PD0

(PCINT17/TXD) PD1

(PCINT18/INT0) PD2

(PCINT19/OC2B/INT1) PD3

(PCINT20/XCK/T0) PD4

VCC

GND

(PCINT6/XTAL1/TOSC1) PB6

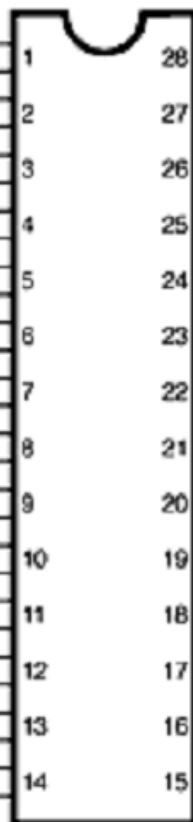
(PCINT7/XTAL2/TOSC2) PB7

(PCINT21/OC0B/T1) PD5

(PCINT22/OC0A/AIN0) PD6

(PCINT23/AIN1) PD7

(PCINT0/CLKO/ICP1) PB0



PC5 (ADC5/SCL/PCINT13)

PC4 (ADC4/SDA/PCINT12)

PC3 (ADC3/PCINT11)

PC2 (ADC2/PCINT10)

PC1 (ADC1/PCINT9)

PC0 (ADC0/PCINT8)

GND

AREF

AVCC

PB5 (SCK/PCINT5)

PB4 (MISO/PCINT4)

PB3 (MOSI/OC2A/PCINT3)

PB2 (SS/OC1B/PCINT2)

PB1 (OC1A/PCINT1)

Arduino function

analog input 5

analog input 4

analog input 3

analog input 2

analog input 1

analog input 0

GND

analog reference

VCC

digital pin 13

digital pin 12

digital pin 11 (PWM)

digital pin 10 (PWM)

digital pin 9 (PWM)

Digital Pins 11, 12 & 13 are used by the ICSP header for MOSI, MISO, SCK connections (Atmega168 pins 17, 18 & 19). Avoid low-impedance loads on these pins when using the ICSP header.